Liver Approach — Harbor Dock

1. Gilbert boat

WHAT YOU SEE

Small rowboat labeled GILBERT with a relaxed man

WHAT IT ENCODES

Gilbert syndrome is the most common inherited cause of unconjugated hyperbilirubinemia (~5-10% of the population), caused by a reduced-promoter UGT1A1 polymorphism (TA insertion, *28 allele) that lowers glucuronosyltransferase activity to ~30%. It is benign and autosomal recessive, producing intermittent mild jaundice without hemolysis or hepatocellular injury (normal AST/ALT, normal reticulocyte count, normal haptoglobin). The relaxed man captures the message: no consequences and no therapy.

BOARD TRAP

WRONG answer: working up Gilbert as hemolysis or liver disease. Discriminator: in Gilbert the LFTs (AST/ALT/ALP) are normal, the elevated bilirubin is unconjugated/indirect, and there is NO anemia, NO elevated reticulocytes, and NORMAL haptoglobin/LDH/smear. Hemolysis would show those abnormalities; Gilbert needs no treatment and no further workup.

2. <3 mg/dL gauge

WHAT YOU SEE

Gauge reading '<3 mg/dL'

WHAT IT ENCODES

Gilbert produces only MILD unconjugated hyperbilirubinemia, typically total bilirubin <3 mg/dL (and unconjugated/indirect predominates, with direct fraction <20%). Levels fluctuate and may rise with triggers but rarely cause overt clinical jaundice. This low ceiling separates Gilbert from Crigler-Najjar, where levels are far higher.

BOARD TRAP

WRONG answer: attributing a bilirubin of 15-40 mg/dL to Gilbert. Discriminator: Gilbert stays <3 mg/dL (occasionally up to ~5 with fasting). Markedly elevated unconjugated bilirubin points to Crigler-Najjar (or brisk hemolysis), not Gilbert.

3. Fasting / stress / illness

WHAT YOU SEE

Icons of fasting, stress, beer/illness over the boat

WHAT IT ENCODES

Gilbert jaundice is precipitated by physiologic stressors that increase bilirubin load or reduce conjugation: fasting/low-calorie intake, dehydration, intercurrent illness or infection, vigorous exercise, menstruation, and alcohol. The classic vignette is a young adult who becomes mildly jaundiced during a viral illness or after fasting/surgery, then normalizes. Recognizing the trigger pattern avoids unnecessary testing.

BOARD TRAP

WRONG answer: ordering imaging or hemolysis labs in a young adult with jaundice that appears only during fasting/illness and resolves spontaneously. Discriminator: a fasting/stress-triggered, self-resolving unconjugated hyperbilirubinemia with otherwise normal labs is Gilbert; reassurance, not workup, is the answer.

4. Reassure only star

WHAT YOU SEE

Star banner 'REASSURE ONLY'

WHAT IT ENCODES

Management of Gilbert syndrome is reassurance only — no drugs, no imaging, no biopsy. It does not progress to chronic liver disease and confers no excess mortality. The single clinically relevant caveat is altered metabolism of UGT1A1-dependent drugs (notably irinotecan/CPT-11).

BOARD TRAP

WRONG answer: starting phenobarbital or UDCA for Gilbert. Discriminator: although phenobarbital can induce UGT1A1 and lower bilirubin, it is not indicated because Gilbert is benign and asymptomatic. The correct management is reassurance and avoidance/dose-adjustment of irinotecan, not pharmacologic treatment.

5. CPT-11 / irinotecan X

WHAT YOU SEE

Red flag 'CPT-11' with skull, crossed out

WHAT IT ENCODES

UGT1A1 deficiency (Gilbert and especially homozygous *28) impairs glucuronidation of SN-38, the active metabolite of irinotecan (CPT-11), causing dangerous accumulation and severe toxicity — profuse diarrhea and life-threatening neutropenia. The crossed-out skull warns to avoid full-dose irinotecan or reduce dosing in UGT1A1-deficient patients (UGT1A1 genotyping can guide dose). This is the chief practical consequence of Gilbert.

BOARD TRAP

WRONG answer: giving standard-dose irinotecan to a UGT1A1-deficient (Gilbert/*28) patient. Discriminator: reduced UGT1A1 activity slows SN-38 detoxification, so the *28/*28 patient is at high risk of severe neutropenia and diarrhea; dose reduction or genotype-guided dosing is required. Atazanavir and indinavir similarly raise bilirubin via UGT1A1 inhibition.

6. ATV inhibits UGT1A1

WHAT YOU SEE

Small banner near CPT-11 flag: 'ATV inhibits UGT1A1'

WHAT IT ENCODES

Atazanavir (ATV), an HIV protease inhibitor, competitively inhibits UGT1A1 and thus causes a benign, reversible indirect (unconjugated) hyperbilirubinemia and scleral icterus — mimicking Gilbert and exaggerated in patients who actually carry the Gilbert polymorphism. Indinavir does the same. Because it is asymptomatic and harmless, jaundice alone is not a reason to stop the drug.

BOARD TRAP

WRONG answer: stopping atazanavir or doing a hepatitis/hemolysis workup because of jaundice. Discriminator: atazanavir-induced jaundice is isolated UNCONJUGATED hyperbilirubinemia with normal transaminases (a UGT1A1 effect, not hepatotoxicity). It is benign and reversible; reassurance, not drug discontinuation, is correct unless transaminases are also rising.

7. MRCP periscope X

WHAT YOU SEE

Crossed-out 'MRCP PERISCOPE'

WHAT IT ENCODES

No biliary imaging (MRCP) is needed for Gilbert or any isolated unconjugated hyperbilirubinemia, because unconjugated jaundice is a problem of bilirubin production or conjugation — not bile-duct obstruction. MRCP is reserved for CONJUGATED hyperbilirubinemia with a cholestatic (ALP-predominant) pattern suggesting mechanical obstruction or PSC.

BOARD TRAP

WRONG answer: ordering MRCP/RUQ ultrasound for isolated indirect hyperbilirubinemia. Discriminator: obstruction causes CONJUGATED hyperbilirubinemia (dark urine, pale stools, high ALP). An unconjugated/indirect picture with normal ALP does not point to the ducts, so imaging is unnecessary.

8. Biopsy grabber X

WHAT YOU SEE

Crossed-out 'BIOPSY GRABBER'

WHAT IT ENCODES

No liver biopsy is needed for Gilbert syndrome; the diagnosis is clinical/biochemical (isolated unconjugated hyperbilirubinemia, normal LFTs, no hemolysis), optionally confirmed by UGT1A1 genotyping. Biopsy would be normal and only adds procedural risk. Tissue is reserved for unexplained hepatocellular or cholestatic disease, not a benign inherited bilirubin transport/conjugation defect.

BOARD TRAP

WRONG answer: biopsying isolated unconjugated hyperbilirubinemia. Discriminator: the inherited hyperbilirubinemias (Gilbert, Crigler-Najjar, Dubin-Johnson, Rotor) are diagnosed by the bilirubin fraction, ancillary tests, and genetics — biopsy is unhelpful (Dubin-Johnson's black liver is incidental, not a diagnostic indication).

9. CN-I boat (skull)

WHAT YOU SEE

Boat labeled CN-I with skull-and-crossbones

WHAT IT ENCODES

Crigler-Najjar type I is the severe autosomal recessive form with COMPLETE absence of UGT1A1 activity, causing massive unconjugated hyperbilirubinemia from birth and a high risk of kernicterus (bilirubin-induced basal ganglia damage). Without treatment it is fatal in infancy; management is intensive phototherapy (and plasmapheresis acutely) as a bridge to definitive liver transplantation, the only cure. The skull-and-crossbones captures its lethality.

BOARD TRAP

WRONG answer: treating CN-I with phenobarbital. Discriminator: CN-I has NO functional enzyme to induce, so phenobarbital does NOT work — that response distinguishes it from CN-II. Definitive therapy is liver transplant; phototherapy/plasmapheresis are temporizing only.

10. 20-45 mg/dL CN-I

WHAT YOU SEE

Gauge '20-45 mg/dL' over CN-I

WHAT IT ENCODES

CN-I produces extremely high unconjugated bilirubin, classically ~20-45 mg/dL, far above Gilbert's <3. These dangerous levels (especially the free, albumin-unbound fraction) cross the immature blood-brain barrier and deposit in the basal ganglia, causing kernicterus. The numeric gap from Gilbert is the fastest discriminator on a vignette.

BOARD TRAP

WRONG answer: ascribing a neonate's bilirubin of 25-40 mg/dL to Gilbert or physiologic jaundice. Discriminator: such extreme unconjugated levels with no enzyme response to phenobarbital indicate CN type I, a transplant-requiring emergency, not benign neonatal or Gilbert jaundice.

11. CN-II boat (check)

WHAT YOU SEE

Boat labeled CN-II with green check

WHAT IT ENCODES

Crigler-Najjar type II (Arias syndrome) has PARTIAL UGT1A1 activity (typically <10% but present), so unconjugated bilirubin is high but lower than CN-I (often ~6-20 mg/dL) and kernicterus is uncommon. It is compatible with a normal lifespan and, unlike CN-I, responds to enzyme induction. The green check signals a livable, treatable condition.

BOARD TRAP

WRONG answer: confusing CN-II with CN-I and recommending transplant, or confusing it with Gilbert. Discriminator: CN-II has intermediate bilirubin levels and, crucially, RESPONDS to phenobarbital (>25% reduction) because residual enzyme exists — neither true of CN-I; levels are higher than Gilbert's <3 mg/dL.

12. >25% decrease (phenobarb)

WHAT YOU SEE

'>25% decrease' label on CN-II

WHAT IT ENCODES

Phenobarbital induces residual UGT1A1, lowering bilirubin by >25% in CN-II — both a diagnostic test and the treatment. Because CN-I has no enzyme to induce, it shows NO response; this phenobarbital challenge cleanly separates the two types. CN-II patients are otherwise managed expectantly with intermittent phenobarbital during flares.

BOARD TRAP

WRONG answer: a phenobarbital response indicates CN-I, or no response indicates CN-II. Discriminator: a >25% bilirubin drop with phenobarbital RULES OUT CN-I and confirms CN-II (residual enzyme present). No response indicates absent enzyme = CN-I, which needs transplant.

13. Dubin-Johnson black boat

WHAT YOU SEE

BLACK sailboat labeled DUBIN-JOHNSON

WHAT IT ENCODES

Dubin-Johnson syndrome is a benign autosomal recessive CONJUGATED (direct) hyperbilirubinemia from a defective canalicular MRP2/ABCC2 transporter that normally exports conjugated bilirubin into bile. Impaired export plus accumulation of melanin-like pigment produces a grossly BLACK liver on biopsy (the black sailboat). It is asymptomatic, needs no treatment, and may worsen with pregnancy or oral contraceptives.

BOARD TRAP

WRONG answer: pursuing an obstructive/cholestatic workup (MRCP, ERCP) for the conjugated hyperbilirubinemia of Dubin-Johnson. Discriminator: in Dubin-Johnson, ALP and bile acids are NORMAL (it is isolated conjugated hyperbilirubinemia, not cholestasis), so there is no obstruction to image. The black liver and gallbladder non-visualization on HIDA are the clues.

14. NOT VISUALIZED

WHAT YOU SEE

'NOT VISUALIZED' over Dubin-Johnson

WHAT IT ENCODES

On HIDA cholescintigraphy in Dubin-Johnson, the defective MRP2 transporter cannot excrete the tracer into bile, so the gallbladder is delayed/NOT visualized (or shows a characteristic delayed, prolonged liver uptake with late gallbladder filling). This functional excretory defect, not duct obstruction, explains the imaging finding and contrasts with Rotor.

BOARD TRAP

WRONG answer: reading the non-visualized gallbladder on HIDA as acute cholecystitis or cystic duct obstruction. Discriminator: in the right clinical setting (chronic benign conjugated hyperbilirubinemia, normal ALP), non-visualization reflects the MRP2 excretory defect of Dubin-Johnson, not an acute obstructive process.

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15. >80% isomer I

WHAT YOU SEE

'>80% ISOMER I' label

WHAT IT ENCODES

Urinary coproporphyrin testing distinguishes Dubin-Johnson from Rotor: TOTAL urinary coproporphyrin is roughly NORMAL in Dubin-Johnson but the fraction that is isomer I is markedly elevated (>80%, vs ~25% normally). In Rotor, total urinary coproporphyrin is high (~2.5-5x normal) with a more modest isomer I proportion. This is the classic exam discriminator between the two conjugated syndromes.

BOARD TRAP

WRONG answer: expecting elevated TOTAL coproporphyrin in Dubin-Johnson. Discriminator: Dubin-Johnson has near-normal TOTAL urinary coproporphyrin with a high isomer-I fraction (>80%); Rotor has markedly elevated TOTAL coproporphyrin. Confusing the total-vs-fraction pattern is the trap.

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16. Rotor boat

WHAT YOU SEE

Sailboat labeled ROTOR, 'VISUALIZED'

WHAT IT ENCODES

Rotor syndrome is a benign autosomal recessive CONJUGATED hyperbilirubinemia caused by defective hepatic re-uptake/storage of bilirubin (loss of OATP1B1 and OATP1B3 transporters), distinct from the MRP2 export defect of Dubin-Johnson. The liver is NORMAL in color (no black pigment), the gallbladder IS visualized on HIDA, and total urinary coproporphyrin is markedly elevated. It is asymptomatic and requires no treatment.

BOARD TRAP

WRONG answer: treating Rotor (or Dubin-Johnson) as obstructive jaundice or as hemolysis. Discriminator: Rotor and Dubin-Johnson are benign conjugated hyperbilirubinemias with NORMAL ALP/bile acids — no obstruction. Rotor is separated from Dubin-Johnson by a normal-colored liver, visualized gallbladder, and high TOTAL coproporphyrin.

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17. Unconj. goat (no collar)

WHAT YOU SEE

White 'free-roaming billy goat without a collar', 'UNCONJ. BILI.'

WHAT IT ENCODES

Unconjugated (indirect) bilirubin is lipophilic and water-INSOLUBLE; it travels tightly bound to albumin (the 'no collar/free-roaming' is a memory hook — it is NOT free in urine because albumin keeps it in the blood). It cannot be filtered by the glomerulus, so it never appears in urine. Causes are prehepatic overproduction (hemolysis, ineffective erythropoiesis) or impaired conjugation (Gilbert, Crigler-Najjar, physiologic neonatal jaundice).

BOARD TRAP

WRONG answer: expecting bilirubinuria with an unconjugated/indirect elevation. Discriminator: because unconjugated bilirubin is albumin-bound and water-insoluble, it is NOT excreted in urine — so isolated indirect hyperbilirubinemia gives NO bilirubinuria. Bilirubin in the urine means a conjugated process.

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18. No bilirubinuria

WHAT YOU SEE

Empty bucket under net, 'NO BILIRUBINURIA'

WHAT IT ENCODES

The empty bucket shows that unconjugated hyperbilirubinemia produces NO bilirubin in the urine, since albumin-bound unconjugated bilirubin cannot be filtered. Conversely, conjugated bilirubin is water-soluble and DOES spill into urine when blood levels rise, giving dark/tea-colored urine (e.g., obstruction, Dubin-Johnson, Rotor, hepatocellular disease). Urine dipstick for bilirubin is therefore a quick fractionation clue.

BOARD TRAP

WRONG answer: dark urine with a positive bilirubin dipstick reflects an unconjugated process. Discriminator: bilirubinuria implies a CONJUGATED (water-soluble) elevation — hepatocellular injury or post-hepatic obstruction — never isolated unconjugated hyperbilirubinemia. Note that urobilinogen, not bilirubin, can rise in hemolysis.

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19. Conj. willy goat

WHAT YOU SEE

Brown 'willy goat', 'CONJ. BILI.'

WHAT IT ENCODES

Conjugated (direct) bilirubin has been glucuronidated by hepatic UGT1A1, making it water-SOLUBLE so it can be excreted into bile and, when retained, into urine. A conjugated-predominant elevation localizes the problem to hepatocellular dysfunction or to bile flow/excretion: intrahepatic cholestasis, the canalicular/transport defects of Dubin-Johnson and Rotor, or extrahepatic (post-hepatic) obstruction.

BOARD TRAP

WRONG answer: lumping all conjugated hyperbilirubinemia as 'obstruction needing ERCP'. Discriminator: conjugated hyperbilirubinemia spans hepatocellular disease, benign transport defects (Dubin-Johnson/Rotor, normal ALP), and obstruction (high ALP, dilated ducts). Check ALP and imaging: normal ALP without ductal dilation points to a transport defect, not a stone or stricture.